FACULTY OF COMPUTING

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SECJ 3553 - ARTIFICIAL INTELLIGENCE

SECTION 02

PROJECT:

SMART AGRICULTURE SYSTEM

THEME:

AGRICULTURE

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1.0 PROPOSAL

PROBEM STATEMENT

Agriculture is one of the biggest contributors to climate changes such as global warming as the activities involved release large amounts of greenhouse gases to the atmosphere. Gases such as methane, nitrous oxide and carbon dioxide are emitted from the use of fossil fuels in agricultural machinery, the open burning of crop leftovers and the breakdown of organic materials in the soil.

Furthermore, the unpredictable climate changes cause irregular weather patterns, extreme temperatures, and prolonged droughts that directly affect agricultural productivity, leading to lower crop yields and economic losses for farmers. Insect pests are also a major threat to agricultural production. These problems make it significantly harder to forecast crop yield, as traditional prediction methods cannot take into account dynamic factors affecting crop productivity, often leading to limited forecasts.

AI SOLUTION

To address these challenges, an AI powered smart farming system can be implemented to monitor environmental conditions, predict crop performance, and support farmers in making more informed decisions. By using drones, IoT soil sensors and satellite data, the system can frequently collect real time data on soil moisture, temperature, nutrient level, pH level and weather conditions. Machine learning (ML) can use its algorithms to analyze these data to detect early signals of the crop such as drought, heat exposure or pest infestation.

In addition, an AI based prediction model can improve the accuracy of crop yield forecasting by considering dynamic climate variables that traditional methods are not capable of. Instead of relying on manual observation, the AI system processes large datasets such as rainfall history, soil quality, humidity, and plant growth patterns. The model then generates precise yield estimates and suggests optimal harvesting schedules and resource allocation strategies.

GOAL OF AI SOLUTIONS

Automation of Irrigation:

This goal focuses on enabling farms to automatically regulate water usage based on the needs of crops and environmental conditions. By ensuring that water is supplied only when necessary, this improves resource efficiency, prevents wastage, and supports sustainable water management, which is very important in areas facing drought or unpredictable rainfall patterns.

Electric and Autonomous Machinery:

The aim here is to reduce reliance on fossil fuels and improve the efficiency of agricultural operations. By using machines that can operate independently and rely on electric power, farms can lower emissions, enhance safety, and perform tasks more consistently. This contributes to more sustainable farming practices and reduces long-term operational costs.

Predictive Crop Modeling:

This goal emphasizes anticipating crop growth and future yield based on current and historical environmental patterns. By forecasting crop performance, farmers can plan ahead, minimize losses, and better prepare for challenges such as climate unpredictability. Accurate predictions support smarter decision-making and improve overall farm productivity.

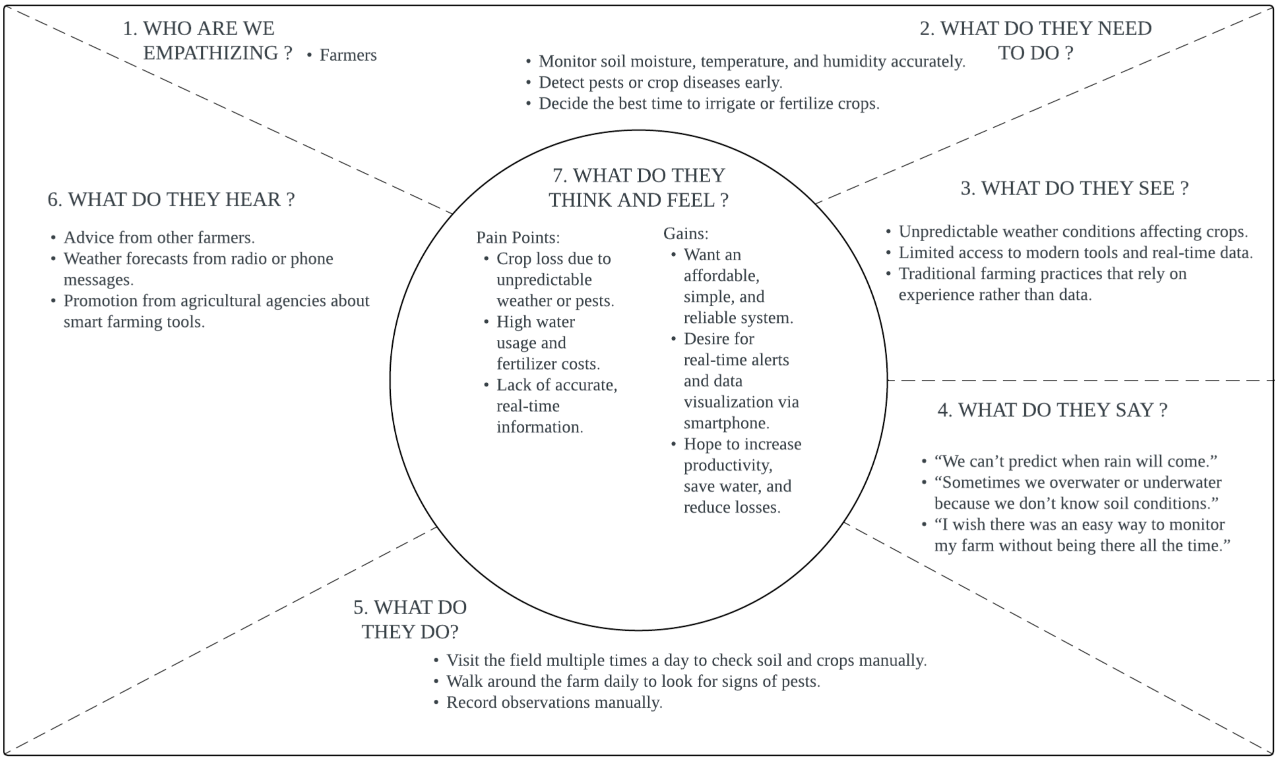
Pest and Disease Prediction:

The objective is to detect potential threats to crops before significant damage occurs. Early identification of pests and diseases helps farmers take preventive action, reduce crop losses, and minimize the use of excessive pesticides. This leads to healthier crops, lower environmental impact, and more stable food production.

Smart Resource Optimization:

This goal focuses on using resources such as water, fertilizer, and energy more effectively. By understanding what crops need at specific times, farmers can avoid overuse or waste of inputs, reducing environmental damage and improving cost efficiency. It promotes sustainable farming practices while maintaining high productivity levels.

PROCESS OF DESIGN THINKING EMPATHY



PROCESS OF DEFINE DESIGN THINKING

USER	NEED	INSIGHT
Farmers who aim to protect their crops, improve productivity, and maintain sustainable agricultural practices even with environmental uncertainties.	Need 1: Farmers need a way to manage pests early to prevent future outbreaks and reduced crop quality. Need 2: Farmers need a way to reduce environmental effects from daily agricultural activity. Need 3: Farmers need a way to increase agricultural productivity even with unpredictable weather changes.	First insight: Farmers struggle to detect early signs of pest activity because manual monitoring is time-consuming and often inaccurate. They need a reliable system that helps them identify pest threats early before they spread and cause major crop damage. Second insight: Many farmers unintentionally contribute to environmental degradation due to limited guidance on sustainable practices. They need support that helps them make environmentally responsible decisions while still maintaining efficient daily farm operations. Third insight: Unpredictable climate conditions make it difficult for farmers to plan and maintain consistent crop productivity. They need better ways to adapt quickly to changing weather to avoid losses and ensure stable agricultural output.

2.0 KNOWLEDGE REPRESENTATION

1. **Automation of Irrigation**

An irrigation system to regulate water usage efficiently, reduce waste, and ensure crops receive the optimal amount of water based on soil condition and weather patterns.

- *IF soil_moisture_level = low AND rain_prediction = false*
  THEN irrigation_status = activate
- *IF10 soil_moisture_level = adequate OR rain_prediction = true*
  THEN irrigation_status = deactivate

Explanation:

This irrigation system uses sensors and weather data to determine whether crops have adequate moisture for proper growth. When the soil moisture is low and no rain is forecasted, the irrigation system activates and waters the plant. When the soil moisture is at a recommended level or rain is forecasted, the irrigation system does not activate to avoid overwatering the plants.

Predicate Logic / First-Order Logic:

Activate Irrigation

$\forall x \forall y (LowMoisture(x) \land \neg RainExpected(y) \rightarrow ActivateIrrigation(x))$

Deactivate Irrigation

$\forall x \forall y (AdequateMoisture(x) \lor RainExpected(t) \rightarrow DeactivateIrrigation(x))$

The first predicate is for activating the irrigation system, where irrigation is activated only when the soil moisture is low and no rain is expected. The second predicate deactivates the irrigation system when the soil has enough moisture or when it is expected to rain.

2. **Electric and Autonomous Machinery**

   Using electric-powered and AI-driven autonomous machines to reduce emissions, improve efficiency, and increase farm safety.

   - *IF machinery_battery_level = low OR terrain_condition = unstable OR sensor_health_status = faulty THEN autonomous_machine_operation = unsafe*
   - *IF obstacle_distance = close AND braking_system_response = slow THEN collision_risk = high*

   Explanation:

   The system utilizes sensors to assess the physical state of a machine in order to maintain safe operations. Low battery, unstable terrain, or faulty sensors mean machines might not be functioning right, so the system calls it unsafe to operate and then calls for protective actions. The AI also scans the environment for obstacles. If an object is too close and the braking response is slow, it detects a high collision risk and slows down or stops the machine. These rules help prevent accidents and ensure reliable autonomous operation.

   Predicate Logic / First-Order Logic:

   1. $\forall x\ (\ \text{lowBattery}(x) \lor \text{unstableTerrain}(x) \lor \text{faultySensor}(x) \rightarrow \text{unsafeOperation}(x)\ )$
   2. $\forall x\ \forall y\ (\ \text{closeObstacle}(x, y) \land \text{slowBraking}(x) \rightarrow \text{highCollisionRisk}(x, y)\ )$

   The first predicate shows that any machine with a low battery, unstable terrain, or faulty sensors is automatically considered unsafe. The second predicate states that if an obstacle is close and braking is slow, the machine is at high risk of collision.

3. Predictive Crop Modeling

This goal emphasizes anticipating crop growth and future yield based on current and historical environmental patterns.

- *IF weather_conditions = extreme OR climate_pattern = unstable THEN yield_prediction = uncertain*
- *IF soil_status = poor OR historical_growth_trend = unstable THEN yield_prediction = low*
- *IF weather_conditions = favourable AND soil_status = good AND historical_growth_trend = stable THEN yield_prediction = high*

Explanation:

When weather conditions are extreme or when climate patterns become unstable, the system most likely produces uncertain yield predictions, which makes planning and resource allocation more challenging. Then, poor soil health or inconsistent past growth trends can also lead to lower yield forecasts. The system can analyze patterns from weather data, soil readings, and historical growth records to automatically predict the outcomes on crop yield.

Predicate Logic/First-Order Logic:

1. $\forall x\ ((\text{ExtremeWeather}(x) \lor \text{UnstableClimate}(x)) \rightarrow \text{Predict}(x, \text{UncertainYield}))$

For every crop instance, if the weather conditions are extreme or the climate pattern is unstable, then the system will predict that the crop yield is uncertain.

2. $\forall x\ ((\text{PoorSoil}(x) \lor \text{DecliningHistory}(x)) \rightarrow \text{Predict}(x, \text{LowYield}))$

For every crop instance, if the soil quality is poor or the historical growth trend is declining, then the system will predict a low crop yield.

3. $\forall x\ ((\text{FavorableWeather}(x) \land \text{GoodSoil}(x) \land \text{StableHistory}(x)) \rightarrow \text{Predict}(x, \text{HighYield}))$

For every crop instance, if the weather is favorable, soil quality is good, and the historical growth trend is stable, then the system will predict a high crop yield.

4. Pest and Disease Prediction

Using AI-enabled sensing and crop-health monitoring to detect early signs of pest and disease threats to reduce pesticide usage and protect crop yield.

- *IF leaf_spot = yes THEN apply_pesticide*
- *IF humidity = high AND temperature = high THEN fungal_removal*

Explanation:

This system uses sensor data to detect early pest and disease risks in the system. If leaf spots are found, pesticide treatment is applied. When both humidity and temperature are high, fungal growth is likely, so fungal removal is performed. This helps reduce pesticide use while protecting crop yield and increasing crop production.

Predicate Logic / First-Order Logic:

$$\forall x\ (LeafSpot(x) \rightarrow ApplyPesticide(x))$$

$$\forall x\ (HumidityHigh(x) \land TemperatureHigh(x) \rightarrow FungalRemoval(x))$$

5. **Smart Resource Optimization**

   Ensuring efficient use of machinery, materials, and energy based on real-time sensor data and predictive AI models.

   - *IF material_wastage_rate = increasing OR remaining_material_stock = low THEN reorder_materials = needed*
   - *IF predicted_material_usage = high AND supplier_delivery_delay = long THEN early_procurement = required*

   Explanation:

   The system helps to minimize wastage and ensures materials are available when needed. If waste goes up or stock levels are too low, it suggests reordering to prevent being out of stock altogether. The system also forecasts material requirements. If it is expected to see a lot of usage, and the deliveries take a long time, it recommends ordering ahead so there are no delays.

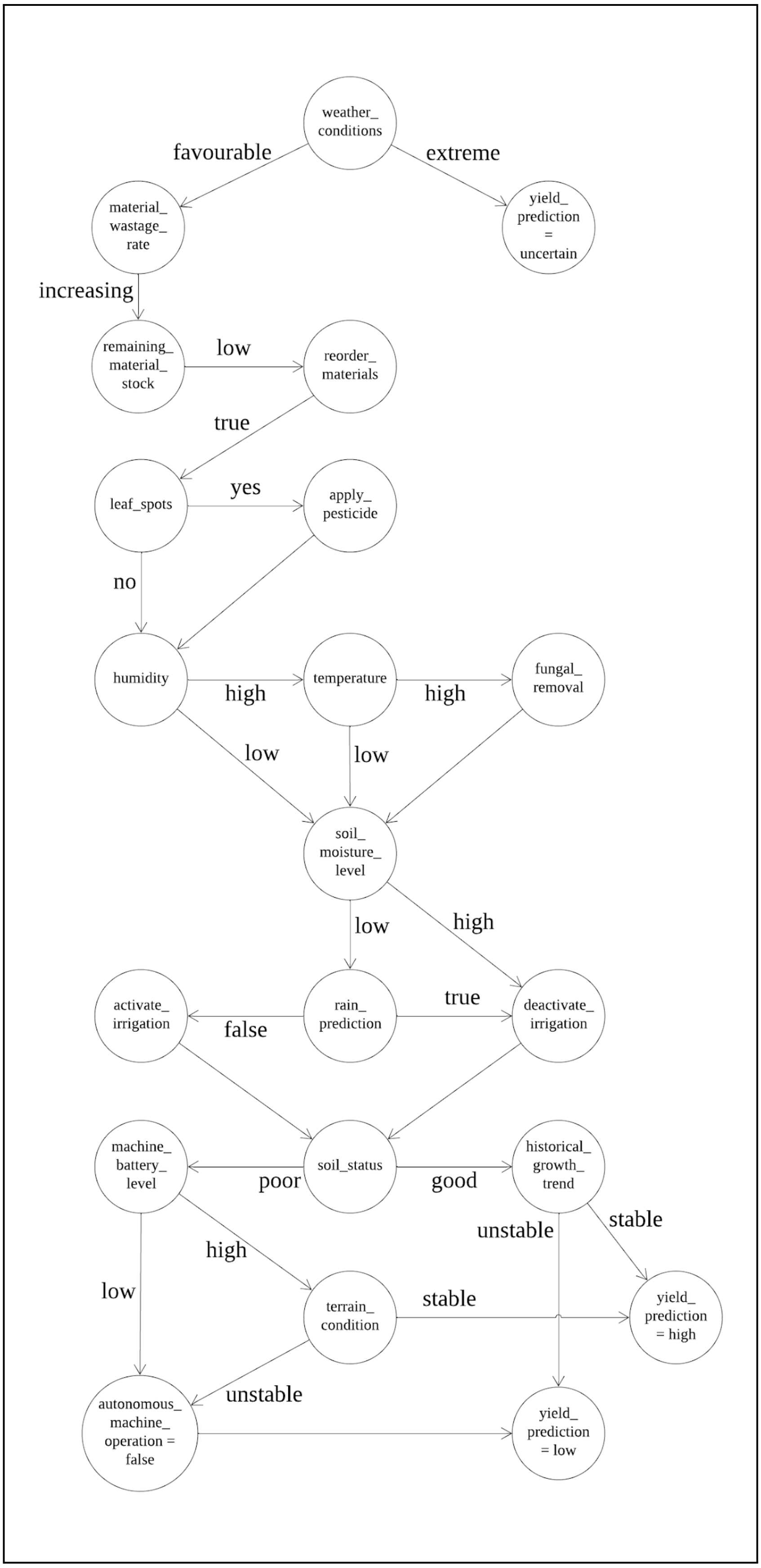
   Predicate Logic / First-Order Logic:

   $\forall x \, (\text{highWastage}(x) \lor \text{lowStock}(x) \rightarrow \text{reorder}(x))$

   $\forall x \, \forall y \, (\text{highUsagePrediction}(x) \land \text{longDelivery}(y) \rightarrow \text{earlyProcurement}(x, y))$

   The first predicate indicates that if a material has high wastage or low stock, the system must reorder it. The second predicate shows that if future demand is high and delivery time is long, early procurement is required.

3.0 PROBLEM FORMULATION BY STATE SPACE GRAPH



Initial State = weather_conditions
Actions = monitoring_material_wastage_rate, check_remaining_material_stock, reorder_materials, check_leaf_spots, apply_pesticide, check_humidity, check_temperature, apply_fungal_removal, check_soil_moisture_level, activate_irrigation, deactivate_irrigation, predict_rain_forecast, check_soil_status, monitoring_machine, battery_level, analyse_historical_growth_trend, check_terrain_condition, predict_yeild, monitoring_autonomous_machine_operation.
Goal = yield_prediction = low **OR** yield_prediction = high **OR** yield_prediction = uncertain
Path Cost = favourable, extreme, increasing, high, low, true, false, yes, no, good, poor, stable, unstable

Description

The system begins by observing the weather conditions. If the weather becomes extreme, the system will classify the crop yield as uncertain. When the conditions are favorable, it continues monitoring wastage rates and material stock. If wastage increases and stock levels become low, the system automatically reorders materials to prevent production delays.

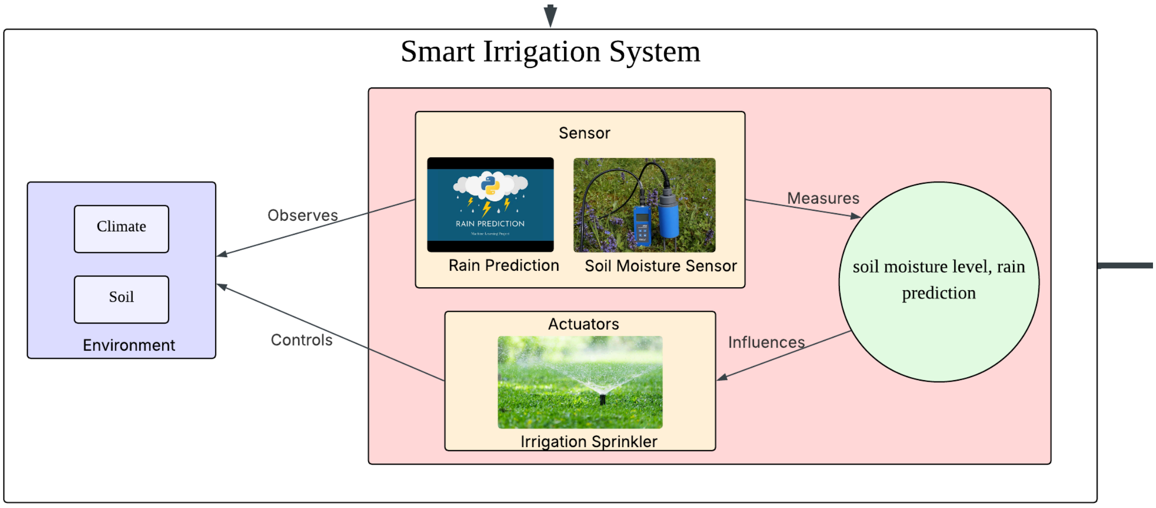
The system also takes care of crop health by detecting leaf spots or high humidity and temperature. When pests or fungal risks are identified, pesticide or fungal treatments are applied to protect the crops. Alongside this, the system monitors soil moisture and rainfall prediction. If the soil is too dry and no rain is expected, irrigation will be activated. Otherwise, it remains off to avoid water waste.

Soil and machinery conditions are also checked to ensure safe and efficient operations. Poor soil quality, low battery levels, or unstable terrain may result in reduced productivity or halting machine operations. Finally, by analyzing soil status and historical growth trends, the system predicts whether the crop yield will be high, low, or uncertain. Through continuous monitoring and smart decision-making, the system ensures optimal resource use and supports better farming outcomes.

4.0 PEAS MODEL FOR SUB-AGENTS AND VISUAL DIAGRAMS

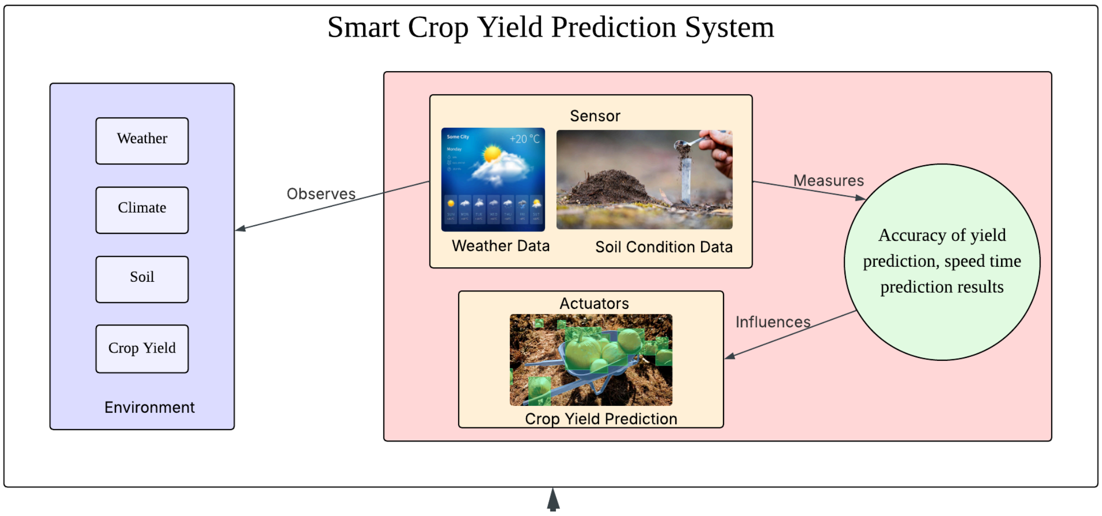
4.1 PEAS Model for Irrigation System

Agent Type	Smart Irrigation System
Performance and Measure	Soil moisture level, rain prediction
Environment	Weather, soil
Actuators	Irrigation sprinkler
Sensors	Soil moisture sensor, weather history data



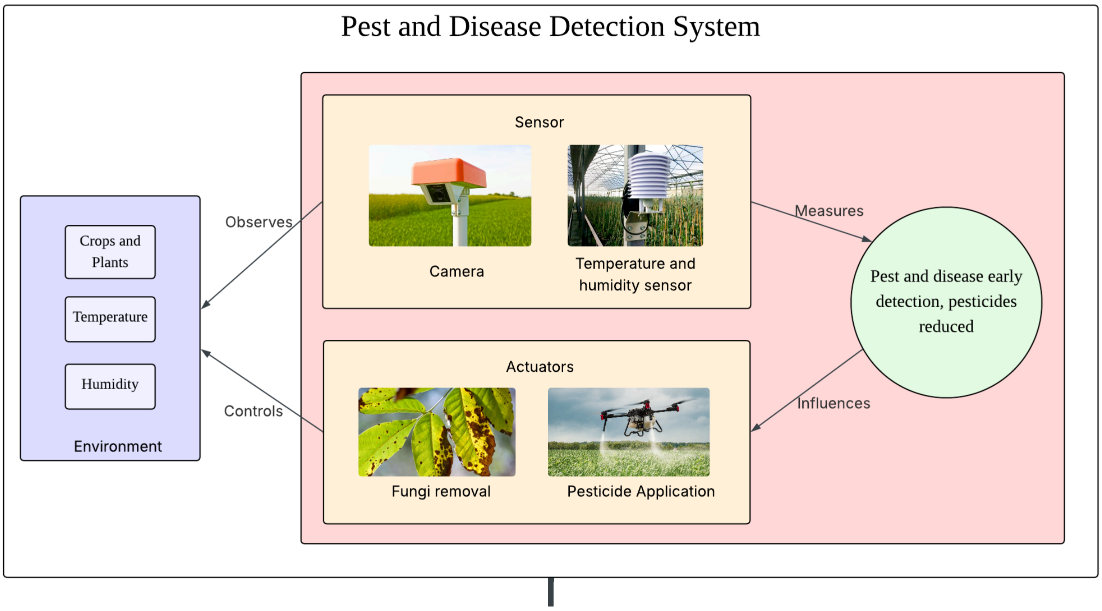
4.2 PEAS Model for Smart Crop Yield Prediction System

Agent Type	Smart Crop Prediction System
Performance and Measure	Accuracy of yield prediction, speed time prediction results
Environment	Weather conditions, climate patterns, soil status, crop growth history
Actuators	Generate prediction of crop yield
Sensors	Weather monitoring data, climate trends, soil condition data, historical performance crop data



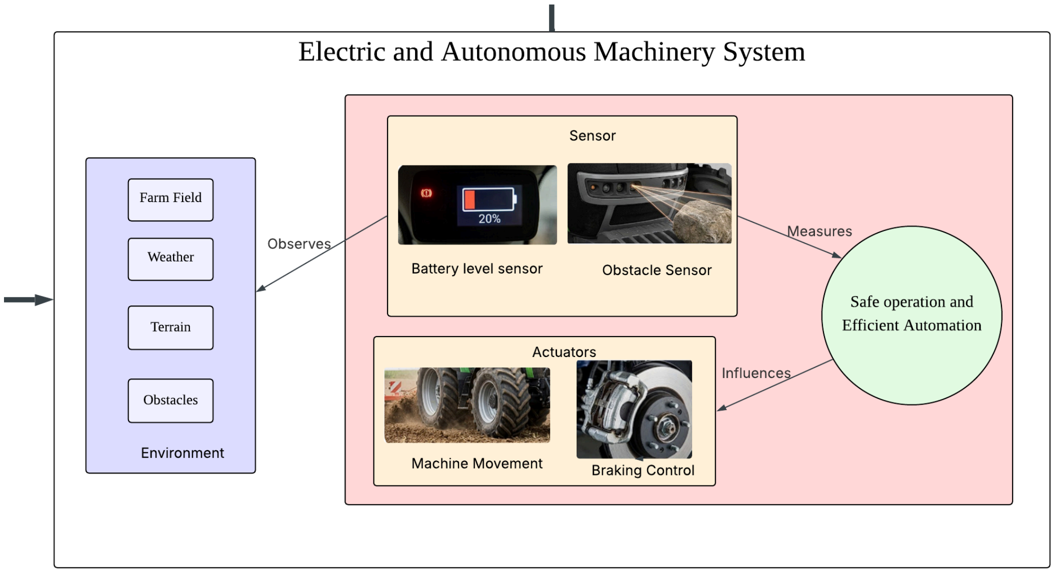
4.3 PEAS Model for Smart Pest and Disease Detection System

Agent Type	Smart Pest and Disease Detection Agent
Performance and Measure	Pest and disease early detection, pesticides reduced
Environment	Crops and plants, temperature, humidity
Actuators	Apply pesticide and fungi removal
Sensors	Camera, temperature sensors, humidity sensors

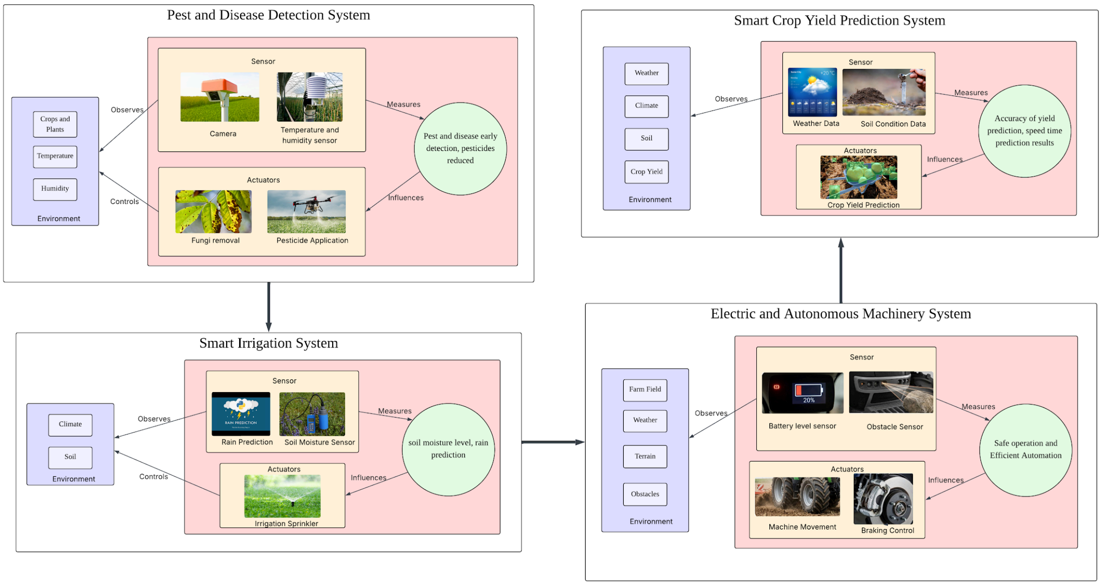


4.4 PEAS Model for Electric and Autonomous Machinery

Agent Type	Electric and Autonomous Machinery
Performance and Measure	Safety, efficiency, collision avoidance, low emissions
Environment	Farm field, weather, soil, terrain, obstacles
Actuators	Machine movement, braking control
Sensors	Battery level sensor, obstacle sensor



4.5 Unified Model Diagram



5.0 PEAS MODEL REPRESENTATION IN PROOF OF CONCEPT (POF)

Performance:

The performance measure serves as the primary goal, which defines success as maximizing agricultural output while reducing the cost of resources like power, water, and pesticides. The precision of autonomous navigation without collisions, the accuracy of disease diagnosis, and the proximity of the expected yield to the actual harvest are all used to quantify success. The prototype guarantees that each action made by the irrigation system or the machinery immediately contributes to the total productivity and sustainability of the farm by aligning these measures.

Environment:

The environment for this system is the physical and digital landscape of the farm. Physically, this includes the kind of soil, the range of crops cultivated, and the unpredictable climate. It also incorporates "dynamic obstacles" that the autonomous machinery must avoid, including farmworkers, wandering animals, or uneven terrain. The yield prediction module must evaluate past climate data and current weather trends from the digital environment.

Actuators:

To interact with and adapt to the surroundings, the system uses a variety of actuators, which are components that do physical or digital work. Mechanical components include the electric motors and steering systems of autonomous tractors, valves that open and close irrigation pipelines, and precision nozzles on drones used for pesticide application and fungal control. On the digital side, the actuators include the user interface or dashboard that operates by displaying the farmer's final crop production forecast. These actuators enable the prototype to transition from passive observation to active intervention, such as shutting off the water when rain is predicted where the weather will provide the required moisture.

Sensors:

The system's sensors, which serve as the data inputs for the whole architecture, are solely responsible for its ability to detect its surroundings. Soil moisture probes and humidity meters give a continuous stream of information on the "soil condition," while high-resolution cameras and optical sensors are utilized for image recognition to identify pests and illnesses early. To guarantee the autonomous machinery functions securely, sensors such as GPS for location tracking and obstacle detectors for obstacle detection manage navigation and safety. Furthermore, the prediction system processes external data that is pulled in via virtual sensors or APIs, such as historical climate records and weather forecasts. When combined, these sensors offer the provided data from the environments that enable the prototype to make informed choices for each of the four modules.

6.0 PROOF OF CONCEPT

Presentation Link: https://youtu.be/7EiY1urxpyQ

Prototype Link: https://ai-smart-agriculture.my.canva.site/